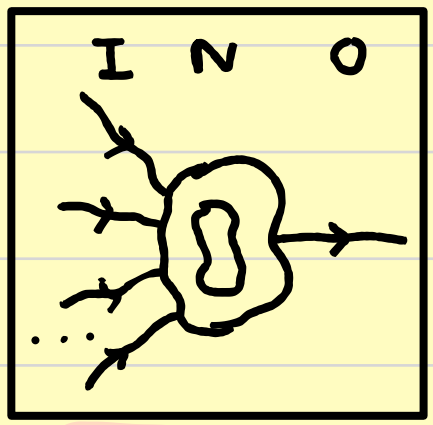
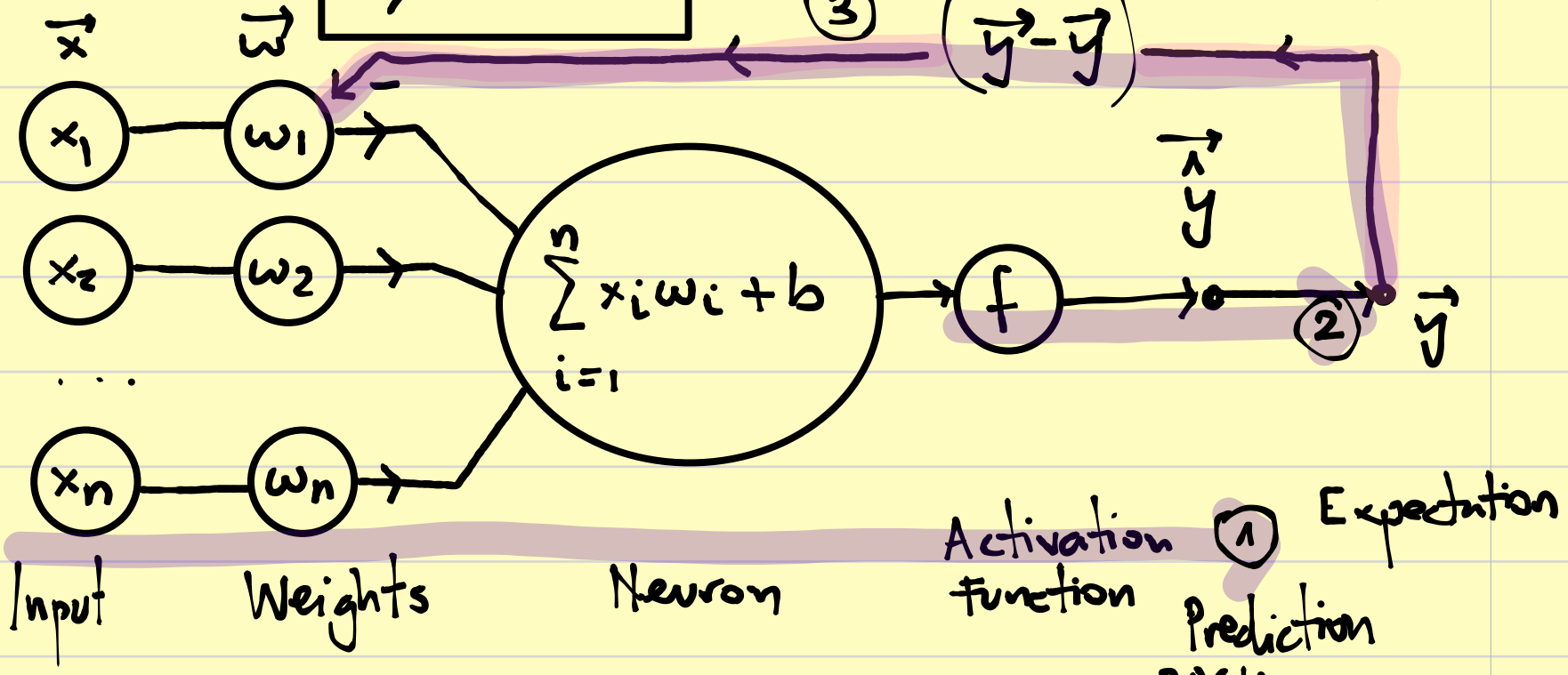
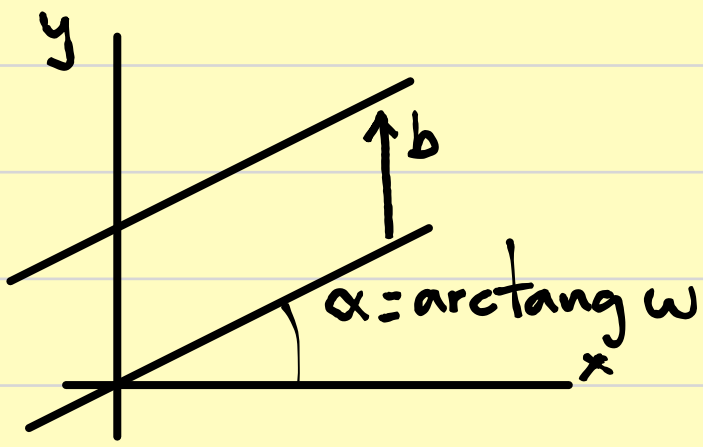
DEEP LEARNING .. by HAND



: PERCEPTRON



① FEED FORWARD ② ACTIVATION FUNCTION ③ BACK PROPAGATION



$y = wx + b \equiv$ LINEAR FUNCTION

- The neuron aggregates $\sum$ all input values x_i with their weights w_i and calculates the weighted sum after the bias.

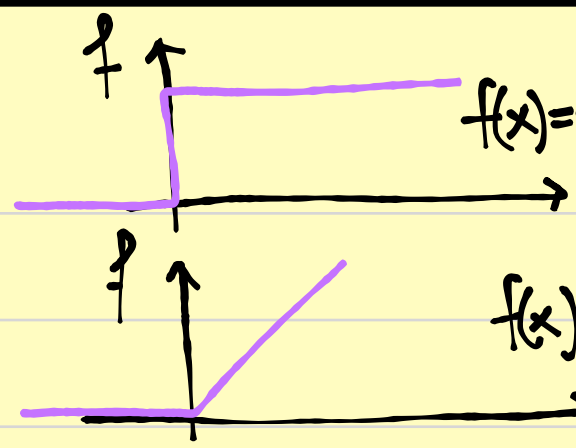
STEP (1) MULTIPLY ALL INPUT VALUES x_i WITH THEIR WEIGHT w_i AND CALCULATE THE WEIGHTED SUM AFTER A BIAS.

STEP (2). APPLY ACTIVATION FUNCTION TO THE ABOVE, GIVING AN OUTPUT OF THE FORM

$$\hat{\vec{y}} = f\left[\sum_{i=1}^{n} x_i w_i + b\right]$$



EXAMPLES OF ACTIVATION FUNCTIONS



$$f(x)=\begin{cases}0 & x\leq 0\\ 1 & x>0\end{cases}$$

STEP FUNCTION

$$f(x)=\begin{cases}0 & x\leq 0\\ x & x>0\end{cases}$$

RECTIFIER LINEAR UNIT (RELU)

STEP (3). WE NEED TO LEARN WHICH WEIGHTS MAKE THE EQUATION $\hat{\vec{y}}=f\left[\sum w_i x_i+b\right]$ AS SIMILAR AS POSSIBLE TO THE EXPECTATION $\vec{y}$. WE WANT TO MINIMIZE THE QUADRATIC ERROR.

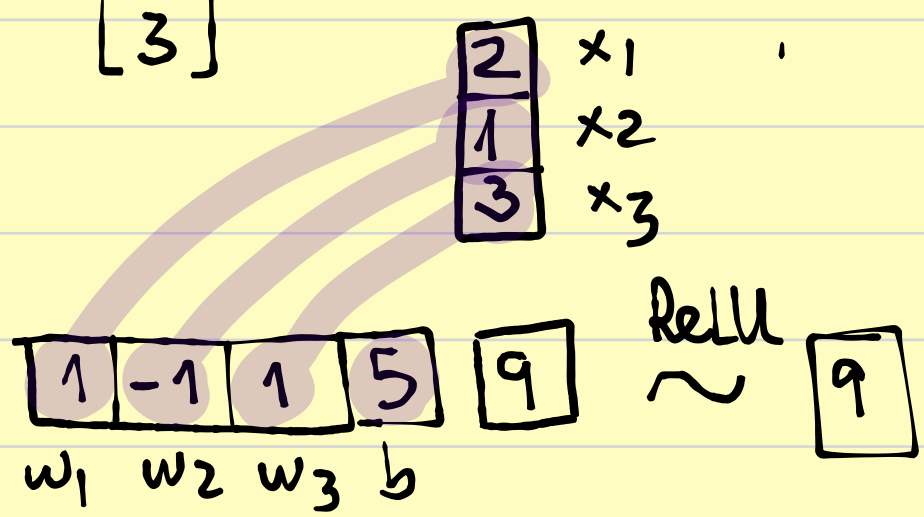
We define a cost function $C=\frac{1}{2}(\vec{y}-\hat{\vec{y}})^2=\frac{1}{2}\left[\vec{y}-f\left[\sum x_i w_i+b\right]\right]^2$

EXAMPLE 1. Single neuron FORWARD PASS.

$b=5$ $\vec{x}=\begin{bmatrix}a\\c\\d\end{bmatrix}$ $\vec{w}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$ $f=$ ReLU OUTPUT?

MATRIXELNR
221378
abcdef

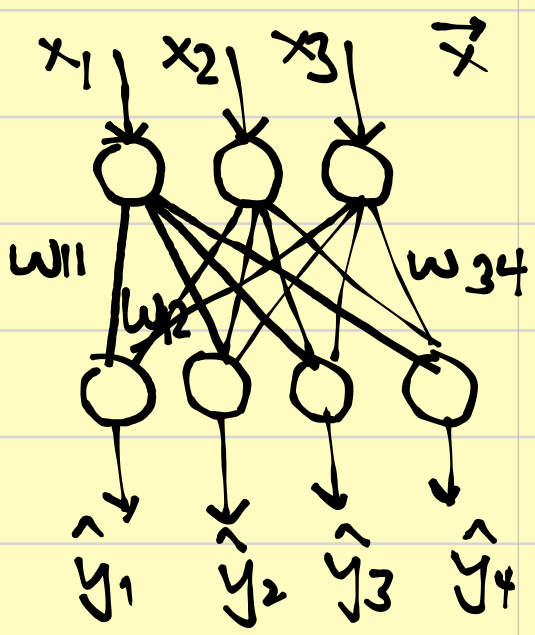
$\vec{x}=\begin{bmatrix}2\\1\\3\end{bmatrix}$ $\vec{x}$



$$\sum_{i=1}^{n} w_i x_i+b=(w_1\cdot x_1+w_2\cdot x_2+w_3\cdot x_3)+b=(1.2+(-1)\cdot 1+1.3)+5=9$$

$$f\left[\sum_{i=1}^{n} w_i x_i+b\right]=\text{ReLU}[9]=9=\hat{y}$$

EXAMPLE 2. TWO LAYERS OF PERCEPTRONS WITH 3 & 4 NEURONS FORWARD PASS



Input vector 3 elements

Weight matrix 12 elements

bias 4 elements

output vector 4 elements

f : ReLU

$\vec{x}$

2
1
3

$W_{3\times4}$ b

·	1	-1	1	5
··	1	1	0	0
∴	0	1	1	1
∷	1	0	1	-2

9
3
5
3

ReLU ~

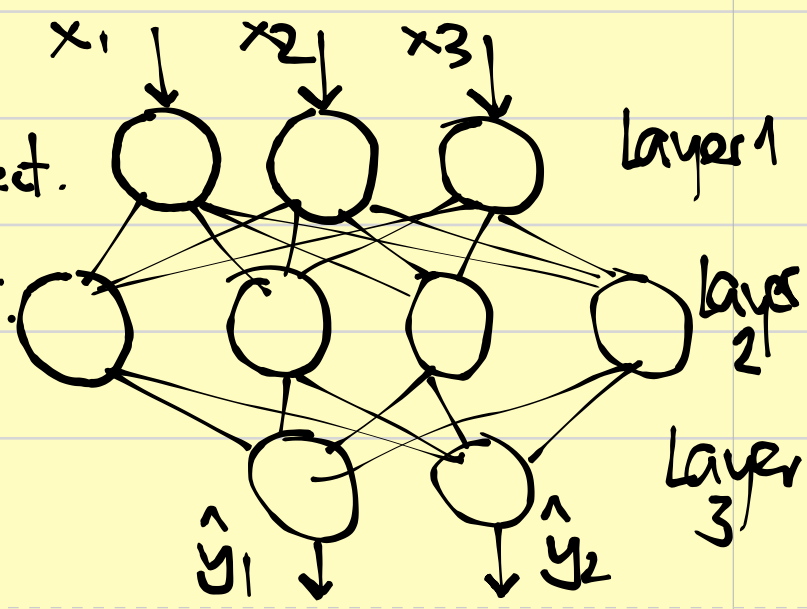
9	$\hat{y}_1$
3	$\hat{y}_2$
5	$\hat{y}_3$
3	$\hat{y}_4$

$$\sum_{i=1}^{n} x_i w_i + b$$

· $2.1+1.(-1)+3.1+5=9$ ∴ $2.0+1.1+3.1+1=5$

·· $2.1+1.1+3.0+0=3$ ∷ $2.1+1.0+3.1+(-2)=3$

EXAMPLE 3. THREE LAYERS OF PERCEPTRONS WITH [3,4,2] NEURONS EACH. FORWARD PASS.



WEIGHT MATRIX L1 & L2 : 12 connect.

" " L2 & L3 : 8 connect.

c
b
a

$\vec{x}$

$\frac{2}{a}\ \frac{1}{b}\ \frac{3}{c}\ \frac{2}{d}\ \frac{6}{e}\ \frac{8}{f}$

$w_{L1 \cdot L2}$ b

1	-1	0	3
0	1	1	5
1	1	0	-2
-1	1	-1	-3

5	ReLU ~	5
8		8
2		2
-7		0

$3.1+1.(-1)+2.0+3=5$

$3.0+1.1+2.1+5=8$

$3.1+1.1+2.0+(-2)=2$

$3(-1)+1.1+2.(-1)+(-3)=-7$

$w_{L2 \cdot L3}$ b

1	-1	1	0	3
-1	1	0	1	-2

2	ReLU ~	2
1		1

$1.5+(-1).8+2.1+0+3=2$

$(-1).5+1.8+0.2+1.0+(-2)=1$

y

GPU. Graphic Powe Unit

EXAMPLE 4 . 4 LAYERS WITH [3,4,2,3] NEURONS WITH FORWARD PASS and BACKPROP.

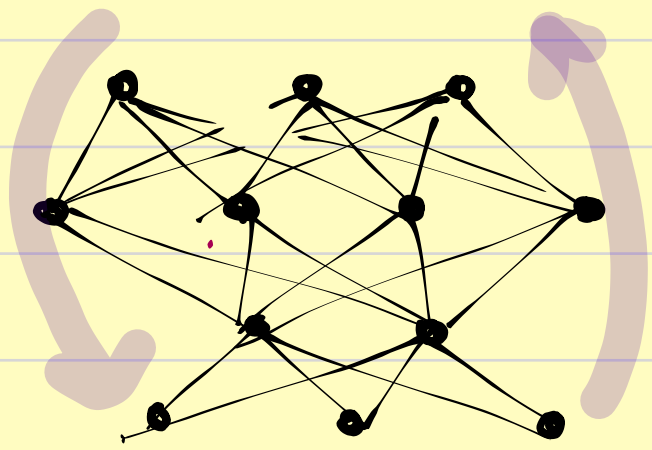
$\vec{x}$

2	1'88	0'12	-2'12	2
1	0'94	0'06	-1'06	1
3	2'82	0'18	3'18	3

$w_{L1 h1}^{*}$

b_1^{*} | 0'94 | 0'06 | -1'06 | 1 |

$w_{h1 h2}^{*}$



L1

h1

h2

L2

$w_{L1 \cdot h1}$ b_1

1	1	-1	0
0	-1	1	1
1	-1	1	3
2	-1	1	-3

0	ReLU ~	0
3		3
7		7
3		3

0	0
-0'18	3
-0'42	7
-0'18	3
-0'06	1

b_1^{*} ERROR1 | 0'94 | 0'06 | -1'06 | 1 |

ERROR2 $\equiv b_2^{*}$

$wh_1 \cdot h_2$

1	-1	1	0	2
1	0	-1	1	-3

b_2

6
-7

ReLU ~

6
0

$wh_2 \cdot L2^*$

-3	0'36	2'64
0	0	0
-0'5	0'06	0'44

(3) b_3^*

-0'6	1

(4) ERROR 2

$wh_2 \cdot L2$

1	-1	2
0	1	-1
1	1	1

b_3

(1)

8
-1
7

Soft Max ~

0'5
0'06
0'44

$\vec{y}$

$-$

1
0
0

$\equiv$

-0'5
0'06
0'44

$\vec{y}$

ERROR 3 (2)

0'5-1 = -0'5
0'06-0 = 0'06
0'44-0 = 0'44

(1)
In the last layer before back propagation, we need elements of the output vector adding up 1. [PROBABILITY DISTR.]

$$\text{SOFTMAX}\begin{bmatrix}8\\-1\\7\end{bmatrix} = \begin{bmatrix}\frac{8}{8+|-1|+7}\\ \frac{|-1|}{8+|-1|+7}\\ \frac{7}{8+|-1|+7}\end{bmatrix} = \begin{bmatrix}\frac{8}{16}\\ \frac{1}{16}\\ \frac{7}{16}\end{bmatrix} = \begin{bmatrix}0'5\\ 0'0625\\ 0'437\end{bmatrix}$$

(2) The error becomes the bias of the previous layer.

(3) We perform an elementwise multiplication of bias & input

6	-3	0'36	2'64
0	0	0	0

$wh_2 \cdot L2^*$

-0'5	0'06	0'44

b_3^*

$6.(-0'5) = -3$ $\quad 6.0'06 = 0'36$ $\quad 6.0'44 = 2'64$

(4) The error for the previous layer is obtained by multiplying the new bias with the old weights

ERROR2	-0'06	1

-0'5	0'06	0'44

b_3^*

W_{h2L2}

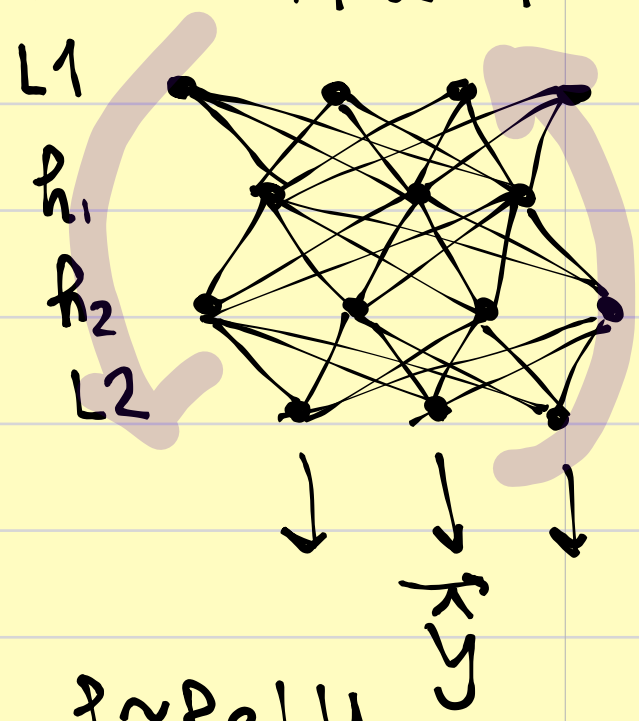
1	-1
0	1
1	1

$-0'5.1 + 0'06.0 + 0'44.1 = -0'06$

$-0'5.(-1) + 0'06.1 + 0'44.1 = 1$

EXAMPLE 5. TO TRAIN. 4 LAYERS WITH $[4,3,4,3]$ NEURONS EACH WITH FF & BP.

$$\vec{x} = \begin{bmatrix} 2 \\ 1 \\ 3 \\ -1 \end{bmatrix}$$



$$W_{L1h1} = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 1 & -1 & 0 & 1 \\ -1 & 0 & 1 & 1 \end{bmatrix} \qquad b_1 = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

$$W_{h1h2} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix} \qquad b_2 = \begin{bmatrix} 0 \\ 1 \\ 3 \\ -3 \end{bmatrix}$$

$f \sim ReLU$

Last layer = Softmax

$$\vec{y} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$w_{h_2L_2} = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \end{bmatrix} \qquad b_3 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

$\vec{x}$

2	-0'38	-0'74	1'46
1	-0'19	-0'37	0'73
3	-0'57	-1'11	2'19
-1	0'19	0'37	-0'73

$w_{L_1h_1}$

-0'19	-0'37	0'73		

b_1^*

$w_{h_1h_2}^*$

$w_{L_1h_1}$

1	1	-1	0	2
1	-1	0	1	3
-1	0	1	1	-1

b_1

2
3
-1

ReLU ~

2
3
0

1'1	0	-0'36	-1'84
1'65	0	-0'54	-2'76
0	0	0	0

0'55	0	-0'18	-0'92

b_1^*

-0'19	-0'37	0'73	

ERROR 1

$w_{h_2L_2}^*$

$w_{h_1h_2}$

1	1	1	0
1	-1	-1	1
-1	0	-1	3
1	1	0	-3

b_2

5
0
1
2

ReLU ~

5	1'87	-2'75	0'9
0	0	0	0
1	0'37	-0'55	0'18
2	0'74	-1'1	0'36

0'37	-0'55	0'18	

b_3^*

b_2^*

0'55	0	-0'18	-0'92

ERROR 2

1	1	0	-1	1
0	-1	0	1	3
1	1	-1	0	-2

$w_{h_2L_2}$

b_3

4
5
2

Softmax ~

0'37
0'45
0'18

$\hat{y}$

$$\text{SoftMax}\begin{bmatrix}4\\5\\2\end{bmatrix}=\begin{bmatrix}\frac{4}{11}\\\frac{5}{11}\\\frac{2}{11}\end{bmatrix}=\begin{bmatrix}0'37\\0'45\\0'18\end{bmatrix}$$

$$\text{ERROR}_3=\begin{bmatrix}0'37\\0'45\\0'18\end{bmatrix}-\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}0'37\\-0'55\\0'18\end{bmatrix}$$